Technical Assignment

Machine Learning / Computer Vision Developer

Company: TrueFox AI Inc.
Role: Machine Learning / Computer Vision Developer/Engineer
Assignment Release Date: Monday, 11 May 2026
Submission Deadline: Sunday, 17 May 2026, 11:59 PM IST
Programming Language: Python

1. Assignment Objective

The candidate must develop a prototype AI-based intelligent camera security system using Python and computer vision/deep learning techniques. The system should analyze video input and detect security-related events in real time or near real time.

The main focus should be on at least **two** of the following video analytics tasks:

1. **Armed person detection**
2. **Abnormal behaviour detection**
3. **Fight or violence detection**
4. **Person detection and tracking**
5. **Suspicious activity detection**

The assignment is intended to evaluate the candidate's ability to design, implement, test, explain, and document a computer vision solution suitable for real-world intelligent surveillance applications.

2. Expected Work

The candidate should develop a Python-based system that accepts image/video/webcam input and performs detection or classification of security-related events.

The solution should include:

- Data collection or dataset selection
- Preprocessing pipeline
- Model selection or model training
- Inference pipeline
- Result visualization
- Performance evaluation
- Clear GitHub documentation
- Explanation of limitations and future improvements

3. Dataset

The candidate may use any publicly available dataset or a small custom dataset.

Possible datasets include:

- Weapon detection datasets
- CCTV abnormal activity datasets
- Violence/fight detection datasets
- Person detection datasets
- Open Images / COCO / Roboflow public datasets

The candidate must clearly mention:

- Dataset source
- Number of images/videos used
- Classes used
- Train/validation/test split
- Any preprocessing or annotation steps

4. Model Requirements

The candidate may use any suitable computer vision model, such as:

- YOLO
- Faster R-CNN
- SSD
- EfficientDet
- CNN/LSTM for video classification
- 3D CNN
- Vision Transformer-based model
- OpenCV-based classical approach, if justified

The candidate may either:

- Train a model from scratch,
- Fine-tune a pretrained model, or
- Use a pretrained model and build a proper inference pipeline.

Use of pretrained models is allowed, but the candidate must clearly explain what was trained, fine-tuned, or reused.

5. Functional Requirements

The submitted system should be able to:

1. Take video input from a file or webcam.
2. Detect at least two security-related events/classes.
3. Draw bounding boxes, labels, confidence scores, or event alerts on the output video.
4. Save output results such as images, videos, logs, or CSV files.
5. Display real-time or near real-time inference results.
6. Provide an alert message when a suspicious/security event is detected.

Example alert:

```
ALERT: Armed person detected with 87% confidence
```

6. Technical Requirements

The project must be coded in Python and should include:

```
main.py
requirements.txt
README.md
models/
results/
data_sample/
notebooks/ or scripts/
```

The candidate should include:

- Clean, modular Python code
- Proper comments
- Error handling where necessary
- Instructions to run the project
- Example input and output
- Model weights or download link
- Result screenshots or videos
- Evaluation metrics

7. Evaluation Metrics

The candidate should report relevant metrics, such as:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- mAP, if object detection is used
- FPS or inference time per frame
- Example success and failure cases

The candidate should also explain:

- False positives
- False negatives
- Model limitations
- How the system can be improved for real deployment

8. GitHub Submission Requirements

The candidate must submit the project through GitHub.

The GitHub repository must include:

1. **README.md** with complete explanation
2. **Source code**
3. **Model file or model download link**
4. **Dataset details**
5. **Training/inference instructions**
6. **Results folder** with screenshots/videos
7. **Evaluation report**
8. **Requirements.txt**
9. **Short technical explanation of the algorithm**

The README should explain:

- Project objective
- System architecture
- Dataset used
- Model used
- Training process
- How to run the code
- Results obtained
- Limitations
- Future improvements

9. Expected GitHub README Structure

```
# AI-Based Intelligent Camera Security System

## 1. Project Overview

## 2. Problem Statement

## 3. Dataset Used

## 4. System Architecture

## 5. Model Selection

## 6. Installation

## 7. How to Run

## 8. Training Details

## 9. Inference Pipeline

## 10. Results

## 11. Evaluation Metrics

## 12. Limitations

## 13. Future Improvements

## 14. References
```

10. Submission Items

The candidate must submit:

- GitHub repository link
- Trained model or model download link
- Results screenshots/videos
- Evaluation report
- Short explanation document or README
- Any assumptions made during development

11. Assessment Criteria

Area	Weight
Problem understanding	10%
Code quality and structure	20%
Model selection and implementation	20%
Result quality and evaluation	20%
Real-time inference pipeline	10%
GitHub documentation	15%
Explanation of limitations and improvements	5%

12. Important Conditions

- The assignment must be completed independently.
- The code must be original or properly referenced.
- Any pretrained model, dataset, or external code must be clearly cited.
- The solution should run on a standard laptop or cloud notebook.
- The candidate should not submit only screenshots without code.
- The candidate should not submit only copied GitHub projects without modification.
- The repository must be public or shared with the recruitment team before the deadline.
- Late submissions may not be considered unless prior approval is given.

13. Final Submission Deadline

Sunday, 17 May 2026, 11:59 PM IST

The candidate should email the GitHub repository link and any model/download links before the deadline.